

Readiness to Train

Project Results

KU Leuven & OH Leuven
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1. Introduction

This report summarises the results of the **Readiness to Train** project: a collaboration between KU Leuven and OH Leuven to build a data-driven system that helps coaching staff optimise daily training intensity for each player. The goal is to answer the question: *"Given everything we know about a player this morning, what training intensity should we prescribe today?"*

The project uses **daily monitoring data** collected by OH Leuven over approximately 20 months (July 2024 – February 2026), covering **28 first-team players** and totalling **14,359 player-day observations**. Four raw datasets were merged into a single analysis-ready file containing 46 engineered features per player-day.

Three experiments were conducted, each answering a different question. Before describing the experiments, we explain the key variables that were constructed during preprocessing.

2. Data and Key Variables

Data sources

Dataset	Rows	Scope	Content
Readiness_Data	14,359	28 players, daily	Wellness z-scores, ACWR, GPS benchmarks, medical status
Raw_Data	9,968	84 players, session-level	Detailed GPS, heart rate, session metadata
Sessions	1,206	Team-level	Match day flags, session types
Games	403	24 players, match-level	High-intensity distance, efforts per ball-in-play, minutes

Data analysis and exploration

Before any modelling, we conducted thorough exploratory data analysis across several notebooks:

- **Raw Data Visualisation** — examined all four datasets individually: variable distributions, missingness patterns, player overlap across datasets (Venn diagrams), temporal coverage, per-player radar charts for wellness and GPS profiles, and cross-dataset correlations.
- **Processed Data Visualisation** — explored the merged RTT.xlsx dataset: feature distributions across the full season, per-player wellness trajectories, ACWR time series, and feature correlation heatmaps to identify redundancy and multicollinearity.

- **Match Analysis** — dedicated exploration of match-level data from Games.xlsx: match intensity distributions, per-player performance profiles, playing time patterns, and the relationship between match load and subsequent wellness responses.
- **Data quality checks** — automated validation of the processed dataset (column completeness, temporal integrity, ACWR flags) and targeted investigation of Training Intensity missingness patterns.

Key engineered variables

Training Intensity Yesterday (the treatment variable): A composite GPS score capturing how hard a player trained on the previous day. Computed as $\tanh(\text{mean}(TD\%, HSD\%, Dec\%, Sprints\%) / 100)$, where each GPS metric is expressed as a percentage of the player's personal match benchmark. The tanh function soft-caps the score in [0, 1). A value of 0 means full rest; values near 1 approach match-equivalent effort. Free days (no training) receive a score of exactly 0.

Match Intensity Yesterday (the match-day outcome): A composite match performance score computed as $\sqrt{HID/BIP \times HIE/BIP} \times \sqrt{\text{clip}(\text{minutes}, 15, 90) / 90}$, where HID/BIP is high-intensity distance per ball-in-play minute and HIE/BIP is high-intensity efforts per ball-in-play minute. This captures both the intensity and duration of a player's match contribution. Only filled on the day after a match.

Status Decrease (the short-term outcome): A binary indicator (0 or 1) capturing whether a player's medical status worsened from one day to the next (e.g., Available → Attention, or Attention → Injured). Prevalence is low: approximately 3–5% of observations.

Why these variables? Training Intensity serves as a proxy for the coaching decision (what load was prescribed), while Match Intensity and Status Decrease serve as outcomes (how the player responded). The causal question is whether we can identify the effect of training intensity on player outcomes while accounting for the fact that coaches assign training based on player state (confounding by indication).

3. Experiment 1 — Match Intensity Prediction

Question: Can we predict how hard a player will perform in the next match, given their morning training state?

Why we tried this: If match-day performance were predictable from training data, we could directly optimise the training regime to maximise match output. This would be the ideal causal target for the Dynamic Treatment Regime framework.

Setup: We predicted Match Intensity Yesterday (continuous) from morning covariates using four models: Ridge Regression, XGBoost, CatBoost, and TabPFN. Each model was tested across 10 lag values (1–10 days of history). The dataset is small — only rows following a match contribute (~248 training, ~72 test samples).

Results

Model	Best Lag	RMSE	Null RMSE	R2	Skill vs Null
TabPFN	8	1.1097	1.3820	0.272	+19.7%
CatBoost	5	1.1932	1.3820	0.158	+13.7%
LinReg	1	1.2250	1.3820	0.113	+11.4%
XGBoost	2	1.2412	1.3820	0.089	+10.2%

The best model (TabPFN) achieves $R^2 = 0.27$, meaning it explains about 27% of the variance in match intensity. While this is better than the null baseline (predicting the mean for everyone), the **per-player analysis reveals the problem:**

- Median per-player $R^2 = -0.75$ (the model is worse than predicting the mean for most players)
- Only **1 out of 17** test-set players has $R^2 > 0$
- The aggregate R^2 of 0.27 is driven by Player ID being the most important feature — some players consistently perform higher than others regardless of training

Adding more historical lags barely helps: the maximum improvement from lag=1 to the best lag is only 1.6%. This confirms that the morning-of-match-day state captures virtually all recoverable signal.

Conclusion

Match-day performance is NOT predictable from training and wellness data alone. Too many unobserved factors intervene: tactical decisions, opponent quality, team composition, player psychology. Since the outcome is not even predictable in a standard (non-causal) ML sense, it will certainly not be identifiable as a causal target either. This validates the decision to focus on more proximal outcomes.

4. Experiment 2 — Training Intensity Prediction

Question: Can we recover the coaching staff's implicit load-assignment policy from observable morning state?

Why this matters: If morning covariates (wellness, fatigue, ACWR, days since/until match) predict the training intensity that coaches actually prescribe, two things follow: (1) the coaching decision-making is **systematic and data-driven**, and (2) the fitted model can serve as a proxy for **player readiness to train** — if the model predicts high intensity, the player is in a state where the coaches would have prescribed a hard session.

Setup: We predicted Training Intensity Yesterday (continuous, [0,1)) from 28 morning-assessment covariates using four models: Ridge Regression, XGBoost, CatBoost, and TabPFN. Each was tested across 3 lag values (1, 2, 3 days of history). The full dataset was used (~14,000 rows after lag creation), giving much larger train and test sets than Experiment 1.

Results

Results across all model-lag combinations show that the coaching policy **is substantially recoverable** from morning covariates. The best models (TabPFN, CatBoost) achieve R² values in the range 0.35–0.45, meaning 35–45% of the variance in prescribed training intensity is explained by observable player state.

Key finding: the coaching staff's load decisions are systematic and largely driven by observable physiological signals. The fitted model can serve as a continuous "Readiness to Train" score for each player-day.

Feature importance analysis (using permutation importance and SHAP) reveals the **top drivers of coaching decisions**:

- **Days Until Match / Days Since Game** — match-cycle position is the single strongest predictor. Coaches follow a clear periodisation pattern: intensity peaks mid-cycle and tapers as the match approaches.
- **ACWR (Acute:Chronic Workload Ratio)** — coaches reduce intensity when ACWR is elevated, consistent with Gabbett's training-injury prevention framework.
- **Previous-day training intensity** — strong autocorrelation indicates coaches plan multi-day loading sequences rather than making independent daily decisions.
- **Wellness (fatigue, readiness, soreness)** — self-reported physical state modulates load prescription, though its importance is secondary to schedule position.

Practical use

The Experiment 2 model can be deployed as a **decision-support tool**: given a player's morning state, the model outputs a predicted training intensity that reflects what the coaching staff would typically prescribe. Deviations between the model's prediction and the actual prescription highlight cases where the coach is making an unusual decision — which may warrant a second look. Additionally, the model serves as a **propensity model** for downstream causal estimation (IPTW weighting).

5. Experiment 3 — Status Decrease Prediction

Question: Can we predict which players are at risk of a status deterioration tomorrow?

What is Status Decrease? A binary indicator (yes/no) capturing whether a player's medical status worsened from one day to the next. For example, going from "Available" to "Attention" or from "Attention" to "Injured" counts as a status decrease. This is a rare event — only about 3–5% of player-days show a status decrease.

Setup: We used Logistic Regression, XGBoost, and CatBoost to predict Status Decrease from morning covariates, running in two modes:

- **Prediction mode** — uses only morning state to predict tomorrow's status (early-warning system)
- **Causal framing mode** — adds Training Intensity as a covariate to examine whether yesterday's load explains today's status change (diagnostic analysis)

Results

Status Decrease is a challenging prediction target due to the severe class imbalance (~97% negative, ~3% positive). Models achieve ROC-AUC values above the 0.5 random baseline but with limited precision at the decision threshold. The causal framing mode reveals an important phenomenon: the coefficient on Training Intensity Yesterday is **negative** in the logistic regression — higher training intensity appears to *reduce* the risk of status decrease.

This counterintuitive result is a textbook example of **confounding by indication**: coaches prescribe higher intensity to players who are already in good condition. The raw association between load and outcome conflates the physiological effect of training with the coach's selection of who receives hard training. Extracting the true causal effect requires the propensity model from Experiment 2 combined with G-computation or IPTW methods.

The negative coefficient on Training Intensity demonstrates exactly why causal methods are needed: naive regression attributes the coach's good judgment (prescribing hard sessions to healthy players) to the training itself.

6. Conclusions and Recommendations

What we achieved

- **Data integration:** Merged four separate monitoring datasets into a unified, analysis-ready file with 46 engineered features per player-day, fully reproducible from raw data.
- **Experiment 1** confirmed that match-day performance is not predictable from training data alone ($R^2 = 0.27$, driven by player identity). This ruled out Match Intensity as a causal target and validated the focus on proximal outcomes.
- **Experiment 2** successfully recovered the coaching staff's load-assignment policy from morning covariates ($R^2 = 0.35\text{--}0.45$). The model can serve as a continuous "Readiness to Train" score and as a propensity model for causal estimation. Feature importance analysis confirmed that periodisation structure (days until match) and workload ratios (ACWR) are the primary drivers of coaching decisions.
- **Experiment 3** built an outcome model for next-day Status Decrease and demonstrated the presence of confounding by indication in the raw data — confirming that causal methods are necessary to isolate the true effect of training load.

Recommendations for OH Leuven

1. **Deploy the Experiment 2 model as a decision-support tool.** The predicted Training Intensity score from Experiment 2 provides a data-driven "expected load" for each player-day. When the actual prescribed load deviates significantly from the prediction, this flags an unusual coaching decision that may warrant review.
2. **Use the Status Decrease model as an early-warning system.** Although the prediction is imperfect (due to low base rate), flagging players with elevated risk before the training session adds a safety layer to the monitoring process.
3. **Continue data collection.** More seasons of data will improve model stability and enable validation across different squad compositions and fixture schedules.
4. **Next research step: causal estimation.** Combining the propensity model (Experiment 2) with the outcome model (Experiment 3) via G-computation or Inverse Probability of Treatment Weighting (IPTW) will allow estimation of the true causal effect of training intensity on player outcomes — free from the confounding by indication demonstrated in Experiment 3.